

Manifold-Informed Architecture Benchmark

PARKINSONS: 2 classes, 22D input

Generated: 2026-04-15 09:41:12 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009) | 2026-04-14 22:20:05
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Best: Standard (176→88) test acc = 0.9333 ± 0.0308

Manifold: 22D → 7D (68.2% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	PARKINSONS
Input dimensionality	22D
Classes	2
Intrinsic dim (d)	7
Variance threshold (τ)	0.9
Epochs	50
Trials	3
Batch size	32
Learning rate	0.001
k neighbors (local PCA)	40

Manifold Discovery

Local PCA over the training set, $k = 40$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 2$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	7.9	0.8	6	9	64.2
0.90	6.3	0.8	4	8	71.5
0.85	5.3	0.7	4	6	75.9
0.80	4.6	0.5	3	5	79.1

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
1	6.4	0.7	4	7
0	5.3	0.4	5	6

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.8974 ± 0.0229	—	—
ManifoldModel ($\tau=0.9$)	0	0.9077 ± 0.0261	—	—
	19,802	0.9333 ± 0.0308	0.1964	0.0471
Standard (176→88) ★				
Manifold (2d→d, d=7)	177	0.8530 ± 0.0697	0.3439	4.8192
Wide Manifold (d+1=8)	202	0.8171 ± 0.0783	0.3542	4.0450
Manifold+ManifoldAdam (d=7)	177	0.8325 ± 0.0591	0.3849	4.7033
PCA→7D + MLP (2d→d)	233	0.8462 ± 0.0477	0.3332	3.6316
Intrinsic Dim (PCA→7D→C)	72	0.8171 ± 0.0799	0.4188	11.3485

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (176→88) — test accuracy 0.9333 ± 0.0308
- **Manifold compression:** 22D → 7D (68.2% of ambient dimensions carry no structure)

Result Figure

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	054030a (-abbrev-re 054030a600978c0e9ffac58faf7157939927d009)
Commit date	2026-04-14 22:20:05 -0400
Commit message	chore(release): bump version to 0.6.0
Python	3.12.13
TensorFlow	2.21.0
Device	CPU (forced)
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
Generated	2026-04-15 09:41:12

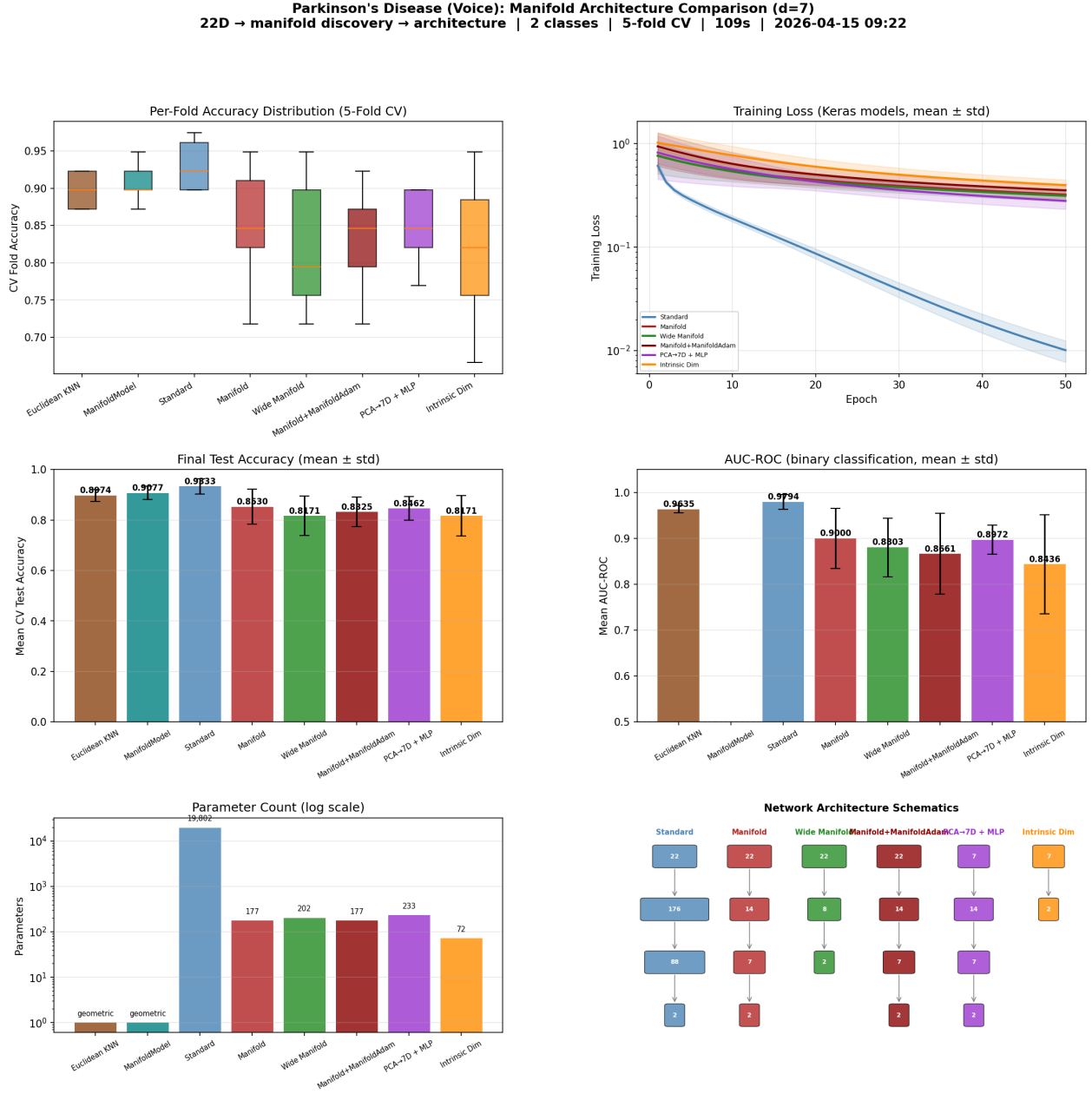


Figure 1: PARKINSONS manifold-informed vs. standard architecture comparison. $d = 7$, 50 epochs, 3 trials.